Chapter 4

Sterilization

Chapter Outline

4.1 Need for Sterilization

4.2 Methods of Sterilization

4.3 Physical Methods of Sterilization

4.4 Sterilization by Heat

4.5 Radiation

4.6 Filtration

EMZHNV



The inoculation loop is sterilized with flame or any other heat source, until it becomes red hot before and after each use.

Learning Objectives

After studying this chapter the student will be able,

- *To understand the concepts of sterilization to maintain aseptic conditions.*
- *To compare the effectiveness of dry heat (red heat, flamimg, incineration, hot air oven), and moist heat (boiling, autoclaving, pasteurization).*
- *To learn the uses of pasteurization in the field of food industry.*
- *To describe the role of radiation in killing pathogens.*
- *To describe how separation of microorganism is achieved through filtration.*

Microorganisms are ubiquitous. They can contaminate, infect or decay inorganic and organic matter. Hence, it becomes necessary to kill or remove them from materials or from areas around us. This is the objective of sterilization.

The process of sterilization is used in Microbiology

- for preventing contamination by extraneous organisms
- in surgery for maintaining asepsis
- in food and drug manufacture for ensuring safety from contaminating organisms

The choice of methods of sterilization depend on the purpose for which it is carried out: the material to be sterilized and the nature of the microorganisms that are to be removed or destroyed.



As early as the stone age, humans used physical methods of microbial control to preserve foods, like drying (desiccation) and salting (osmotic pressure).